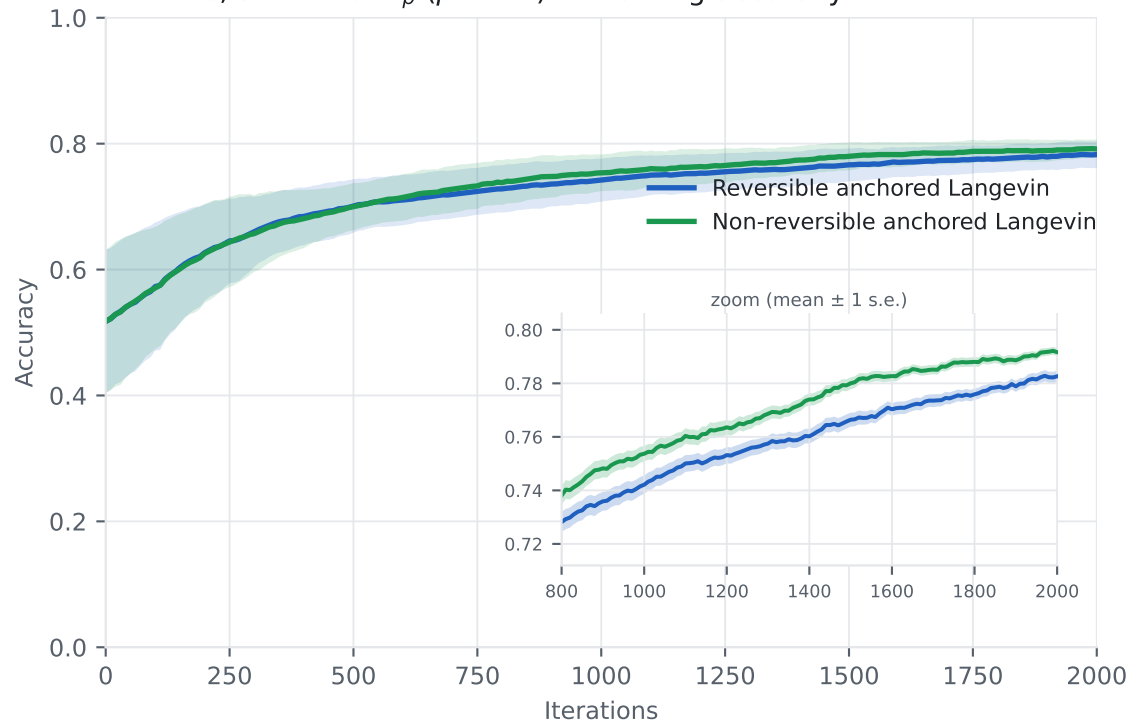
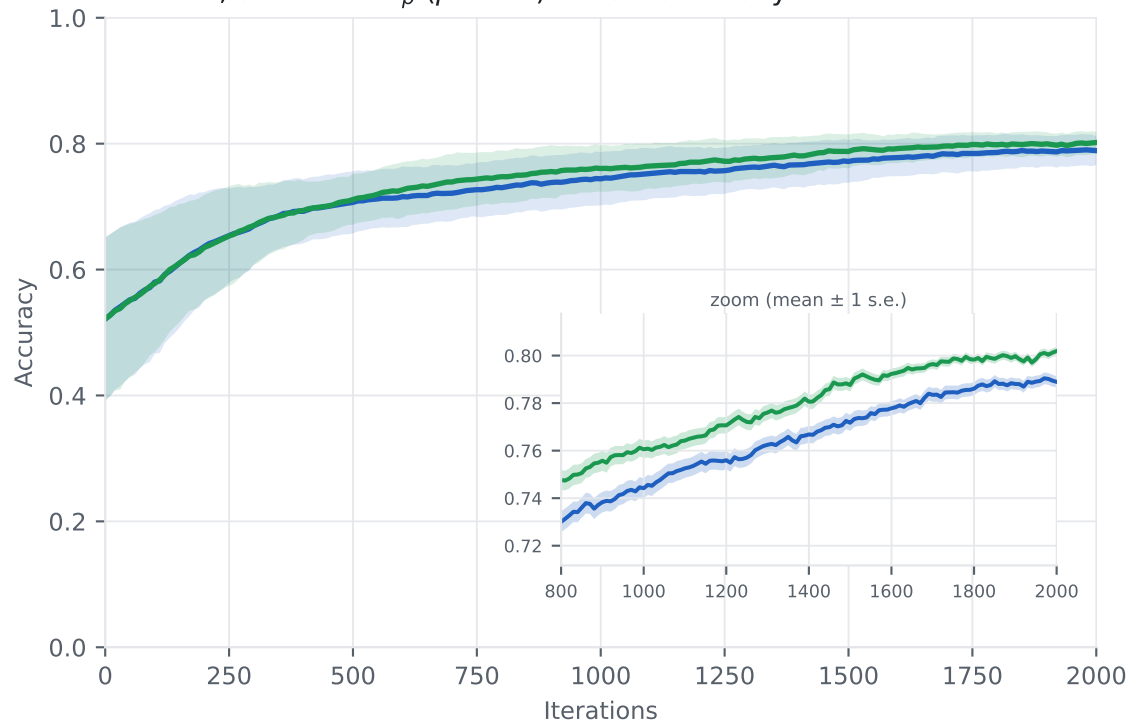


titanic_lp: exact gradient, LASSO anchor

TITANIC, smoothed ℓ_p ($p = 2.4$) — training accuracy



TITANIC, smoothed ℓ_p ($p = 2.4$) — test accuracy



TITANIC -- smoothed ℓ_p , $g(w) = \sum_i (w_i^2 + \text{eps}^2)^{p/2} \leq \text{Lambda}$, $p = 2.4$, $\text{eps} = 0.18$, $\text{Lambda} = 4$. $d = 10$ (intercept w_0 + nine features), no mini-batching: the EXACT gradient is used at every step. $U = \sum \text{softplus}(-y x \cdot w + w_0^2 / (2 \sigma^2)) + \text{lambda} \sum_{j \geq 1} |w_j|$; U_0 replaces $|w_j|$ by $\sqrt{w_j^2 + \delta^2}$; $a = \exp(U - U_0)$. $\sigma = 10$, $\text{lambda} = 7.12 = 0.01$. $n_{\text{train}}, \delta = 0.01082$, so a in $[0.50, 1]$. Block strengths $s = (5, 5, 5)$ on $w_1..w_9$ only. Step $\eta = 1e-05$. $R = 100$ coupled replicates (same W_0 , same Gaussian stream) on one fixed stratified 80/20 split ($n_{\text{train}} = 712$, $n_{\text{test}} = 179$). Paired final-iterate difference, non-reversible minus reversible: training $+0.0090$ ($t = +4.72$), test $+0.0130$ ($t = +5.51$). Bands: mean $\pm$ 1 SD (ddof = 1) across replicates -- repeat-run variability at a fixed split, not a confidence or credible interval.